The Language of Business analytics

Foundational Concepts of Analytics

① Business analytics: Is the use of data, statical methods and models to improve business decisions and performance. It goes beyond reporting by identifying opportunities for efficiency and growth. e.g A retailer using analytics to improve inventory and reduce stockouts

② Data Analytics: Is an act of assessing raw datasets to reveal hidden insights, trends and relationships. It is applicable to almost all industries from healthcare to finance e.g Heath care data analyst assessing and analyzing disease survieillence data to detect outbreaks.

③ Business Intelligence (BI): It is a package of Data systems and tools used to collect analyze and visualize data in support of operational and strategic decisions. It is design for reporting and dashboards rather than predictive modelling: eg Dashboard showing ~~most~~ business performance by period and geography (location)

④ Data Driven Decision making: Application of insights derived from data analysis to guide operational and strategic decision rather than intuition.

It improves accountability and reduces bias eg loan approvals by banks is based on credit score and repayment history.

(5) Big data: These are large dataset defined by their volume, velocity and variety. Advanced data tools and methods are used to process + analyze these datasets eg processing + analyzing unstructured data sets such as social media post.

(6) Descriptive analytics: This is a type of analytics that summarise the past events, answering the 'what' question ('what happened'). It gives a baseline understanding of performance by turning raw data into reports and dashboards eg monthly revenue trends by regions.

(7) Diagnostic Analysis: This is a type of analytics that answers the deep question like why + how something happened. It identifies causes and correlations behind the the performance change e.g Analyzing why and how sales dropped in one region by looking at the competitor activity or pricing shift.

(8) Predictive Analysis: This is the type of analytics that uses statistical models + machine learning to ~~fore~~ forecast

Future outcomes. It helps business anticipate ~~future~~ risk and opportunities: eg predicting customer churn base on past purchase behavior and engagement levels

(9) Prescriptive analytics: This is a types of analytics that prescribe an action to achieve the desired outcomes. It use optimization and simulation models to recomend strategies e.g recommend targeted discounts to retain at risk customers.

(10) Data liracy: Refers to an ability to to read, interpret and communicate with data effectively. It is an important skill for managers and employees in the digital economy e.g being able to correctly interprete dashboard to guide business strategy.

~~(11) Structured data~~: Data, Measurement + Reporting.

(11) Structured data: It is a data that is organized in rows and colums, making it easy to store and query: eg customer transaction records in excel.

(12) Unstructured data: Data without fixed format, such as text, images or videos eg customers reviews on social media

(13) Data warehouse: Central repository that integrates and stores cleaned data from multiple sources for analysis e.g storing company wide sales

(14) Data mining: It is a process of uncovering hidden patterns and relationships in large datasets. e.g Finding that weekend are consistently higher than weekday sales

(15) Data visualization: This is a process of showing data visually through charts, graphs and dashboards to make insights easier to understand e.g. A map shows geographies (regions) with highest revenue.

(16) Dashboard: Real time view of KPIs + metrics normally it is interactive e.g PowerBI dashboard daily sales and customer activity

(17) Metric: It is a measure value used to track performance e.g websites visits per day.

(18) Key performance Indicators: It is a metric linked directly to strategic goals and decision making e.g. Conversion Rate of an e-commerce site.

(19) Benchmark: It is a reference point used to compare performance against industry standards or competitors e.g Comparing sales against the average performance in a certain industry.

(20) Segmentation: Is to divied data (eg customer data) into groups based on shared traits such as demography or behavior eg ~~split~~ dividing customers into groups to tailor marketing strategy

Customer + Sales KPIs

(21) Conversion Rate: % of visitors of the website who take desired action, such as making purchase eg 5% of website visitors buy product.

(22) Customer acquisition cost (CA): It is a cost of gaining one new customer, including marketing and sales expenses e.g spending R200 ~~[illegible]~~ in advert to get one buyer

(23) Customer lifetime value (CLV/LTV): It is a total profit expected from a customer over their relationship with the company e.g A subscriber worth R10 000 over 5 years.

(24) Churn Rate: This is % of customers who stop using the service during a given period eg 10% of subscribers cancel monthly

(25) Customer Retention Rate: % of customer who continue to use the service 90% of users renew subscriptions

(26) Net promoter Score (NPS): It is a measure of customer loyalty through asking if they would recommend the company. eg Score above 40 shows strong loyalty

(27) Average order value: It is the average revenue per transaction eg customer spend R500.00 per order.

(28) Sales funnel: It is stages customers move through from awareness to purchase eg leads moving from ad clicks to checkouts.

Marketing + Campaign Analytics

(29) Marketing campaign: It is a coordinated set of marketing actions aimed at achieving specific goals eg holiday discount campaign.

(30) Post Campaign Analysis: It is a process of reviewing campaign outcomes to measure effectiveness. eg Checking if ads boost sales

(31) Incremental Revenue: This is an extra revenue generated by a campaign beyond the baseline eg R2000 more sales after ads

(32) Return on Investment (ROI): This is a profit compared to cost, it is a classic measure of efficiency eg R5 earned for every R2 spent

(33) Return on Ad Spend (ROAS): This is a revenue generated per advertising money spent eg R10 revenue per R2 ad spend

(34) Click through Rate: This is a % of viewers who click on the link or ad e.g 5% of email ~~recip~~ recipients click a link

(35) Bounce Rate: This is % of visitors who leave site quickly without engaging e.g 40% leave after one page

(36) A/B Testing: This is comparing two versions of a campaign element to see which performs better. eg testing two ad headlines.

(37) lead generation: This refers to capturing potential customer's interest and contact details eg collecting emails via signup forms

(38) Marketing attribution: This is to identify which channel or campaign drove conversion eg email campaign credited for conversions

Reference list

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